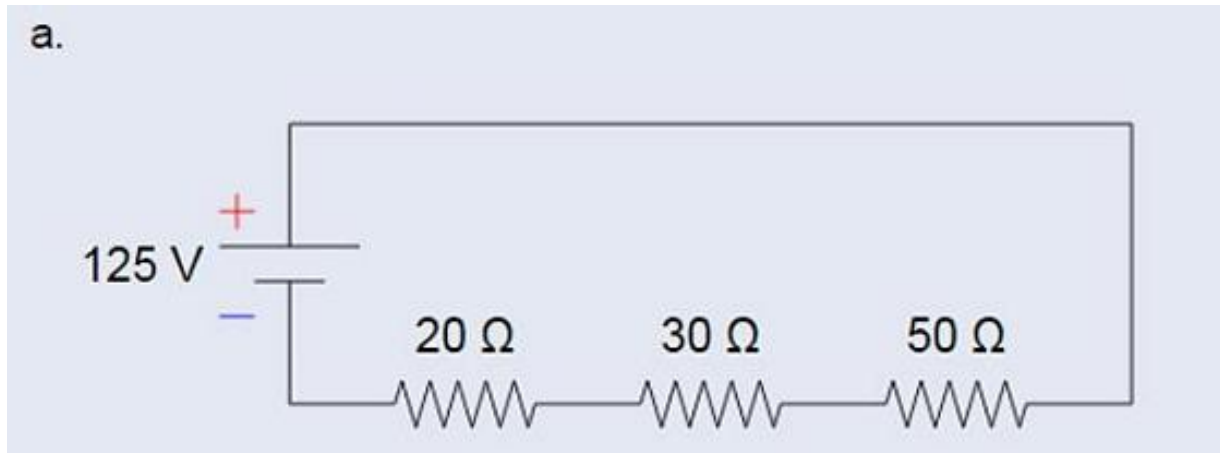


ICS 4111: Embedded Systems & IoT

Practical Exercise 2: Circuit Design

Part A



1. Calculate the total resistance in ohms;

$$R_{\text{total}} = R_1 + R_2 + R_3$$

$$R_{\text{total}} = 20 + 30 + 50$$

$$R_{\text{total}} = \mathbf{100\ \Omega}$$

2. Calculate the total current from the power supply;

$$I = \frac{V}{R} = \frac{125}{100} = \mathbf{1.25\ A}$$

3. Calculate the current through each resistor;

$$I_1 = I_2 = I_3 = \mathbf{1.25\ A}$$

4. Calculate the voltage drop through each resistor;

$$V = IR$$

$$V_1 = 1.25 \times 20 = \mathbf{25\ V}$$

$$V_2 = 1.25 \times 30 = \mathbf{37.5\ V}$$

$$V_3 = 1.25 \times 50 = \mathbf{62.5\ V}$$

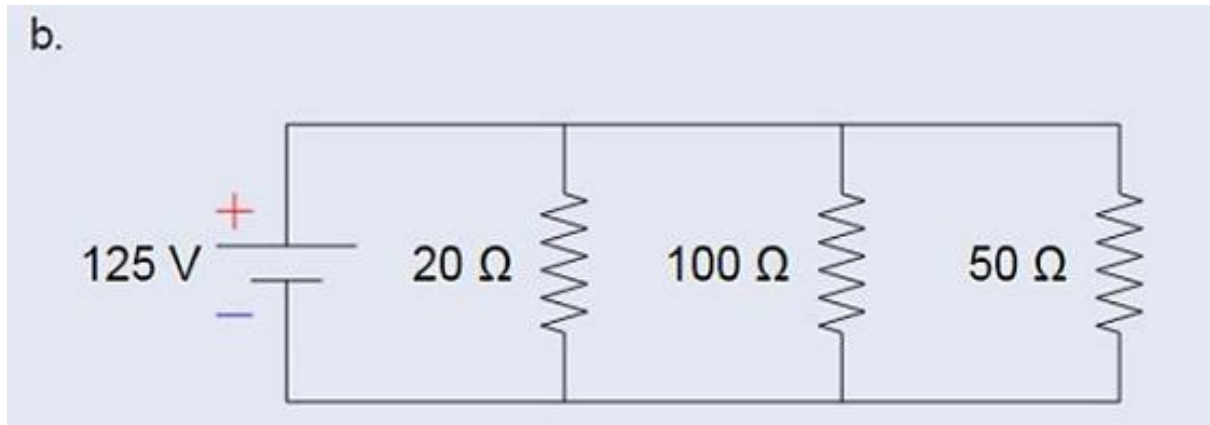
5. Calculate the power dissipated through each resistor (use Watt's Law: Power = Voltage * Current);

$$P = V \times I$$

$$P_1 = 25 \times 1.25 = \mathbf{31.25 \text{ W}}$$

$$P_2 = 37.5 \times 1.25 = \mathbf{46.875 \text{ W}}$$

$$P_3 = 62.5 \times 1.25 = \mathbf{78.125 \text{ W}}$$



1. Calculate the total resistance in ohms;

$$1/R_{\text{total}} = 1/R_1 + 1/R_2 + 1/R_3$$

$$1/R_{\text{total}} = 1/20 + 1/100 + 1/50$$

$$1/R_{\text{total}} = 2/25; R_{\text{total}} = 25/2 = \mathbf{12.5 \Omega}$$

2. Calculate the total current from the power supply;

$$I = \frac{V}{R} = \frac{125}{12.5} = \mathbf{10 \text{ A}}$$

3. Calculate the current through each resistor;

$$I_1 = \frac{V}{R} = \frac{125}{20} = \mathbf{6.25 \text{ A}}$$

$$I_2 = \frac{V}{R} = \frac{125}{100} = \mathbf{1.25 \text{ A}}$$

$$I_3 = \frac{V}{R} = \frac{125}{50} = \mathbf{2.5 \text{ A}}$$

4. Calculate the voltage drop through each resistor;

$$V_1 = V_2 = V_3 = \mathbf{125 \text{ V}}$$

5. Calculate the power dissipated through each resistor (use Watt's Law: Power = Voltage * Current);

$$P = V \times I$$

$$P_1 = 125 \times 6.25 = \mathbf{781.25 \text{ W}}$$

$$P_2 = 125 \times 1.25 = \mathbf{156.25 \text{ W}}$$

$$P_3 = 125 \times 2.5 = \mathbf{312.5 \text{ W}}$$